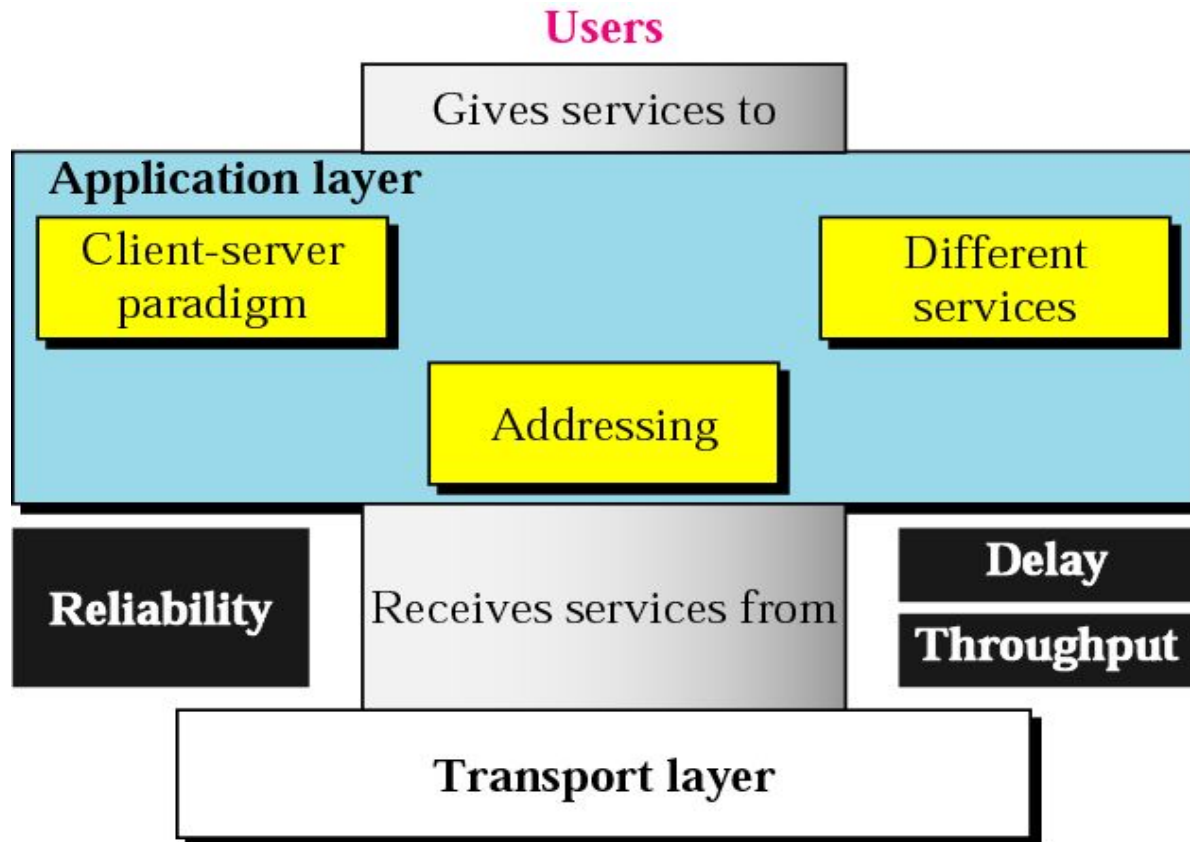


Computer Communication & Networks

Lecture 27

Application Layer: Electronic mail and FTP

Application Layer



Application Layer Topics to Cover

DNS

Remote Logging

Electronic Mail

File Transfer

-
- One of the most popular Internet services is electronic mail (e-mail). The designers of the Internet probably never imagined the popularity of this application program. Its architecture consists of several components that we discuss in this chapter.

Electronic Mail

SMTP, POP3, IMAP

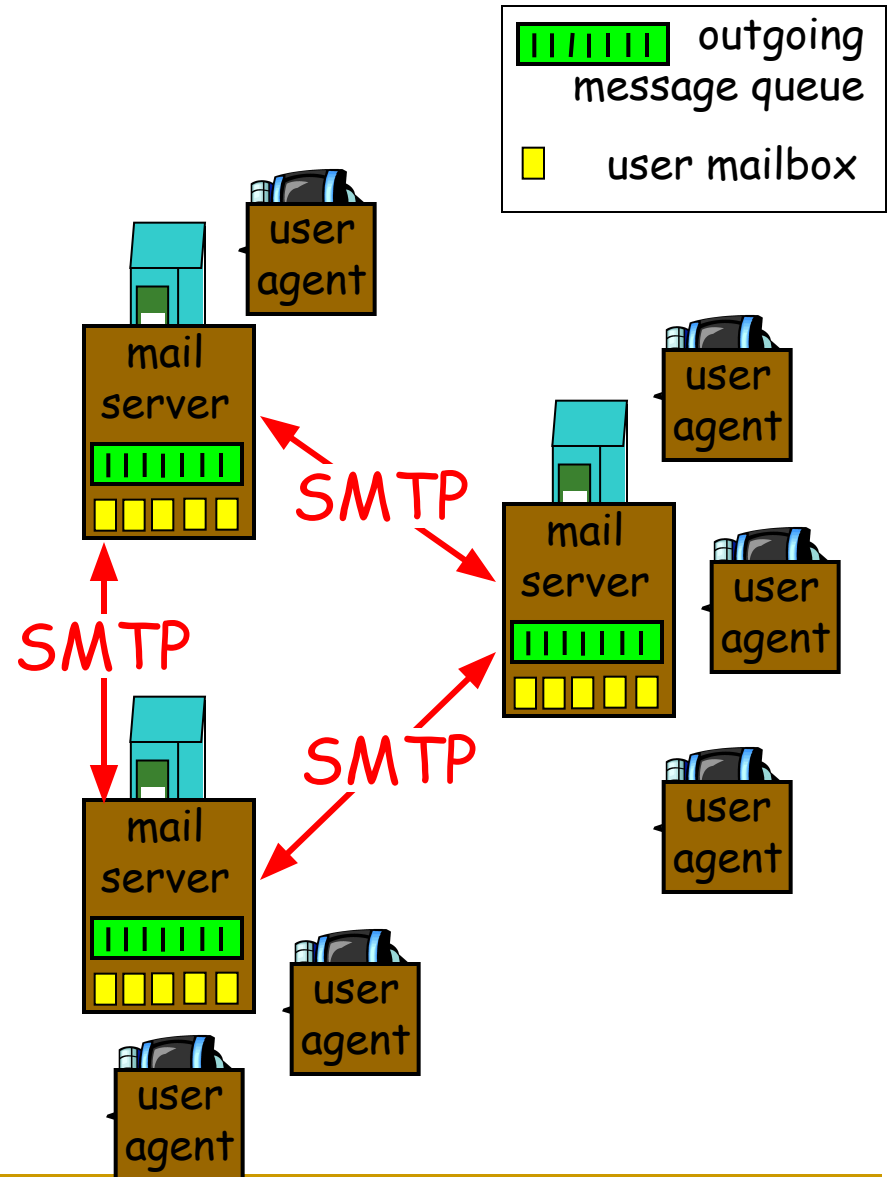
Electronic Mail

Three major components:

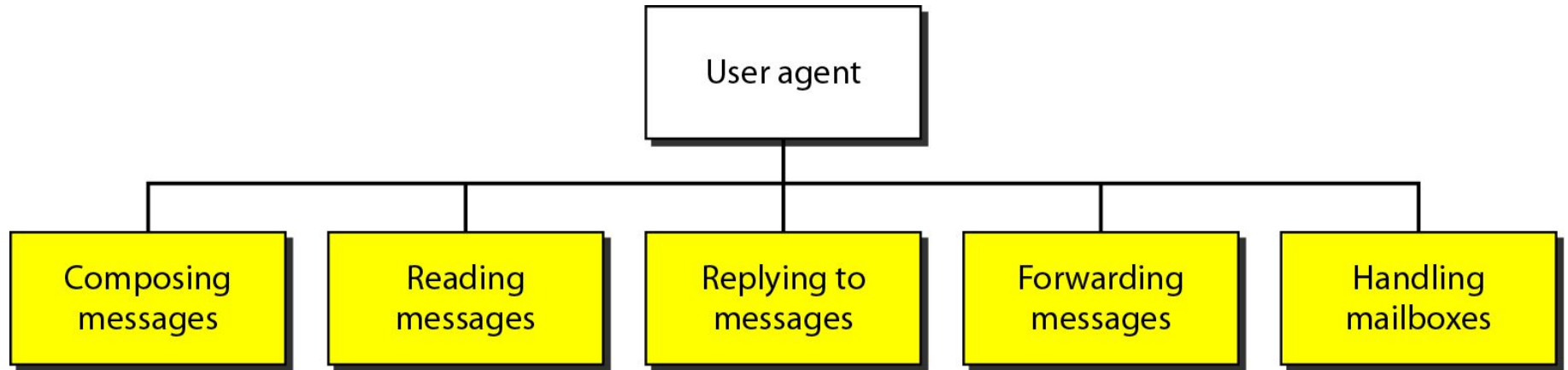
- user agents
- mail servers
- simple mail transfer protocol: SMTP

User Agent

- a.k.a. “mail reader”
- composing, editing, reading mail messages
- e.g., Eudora, Outlook, elm, Netscape Messenger
- outgoing, incoming messages stored on server



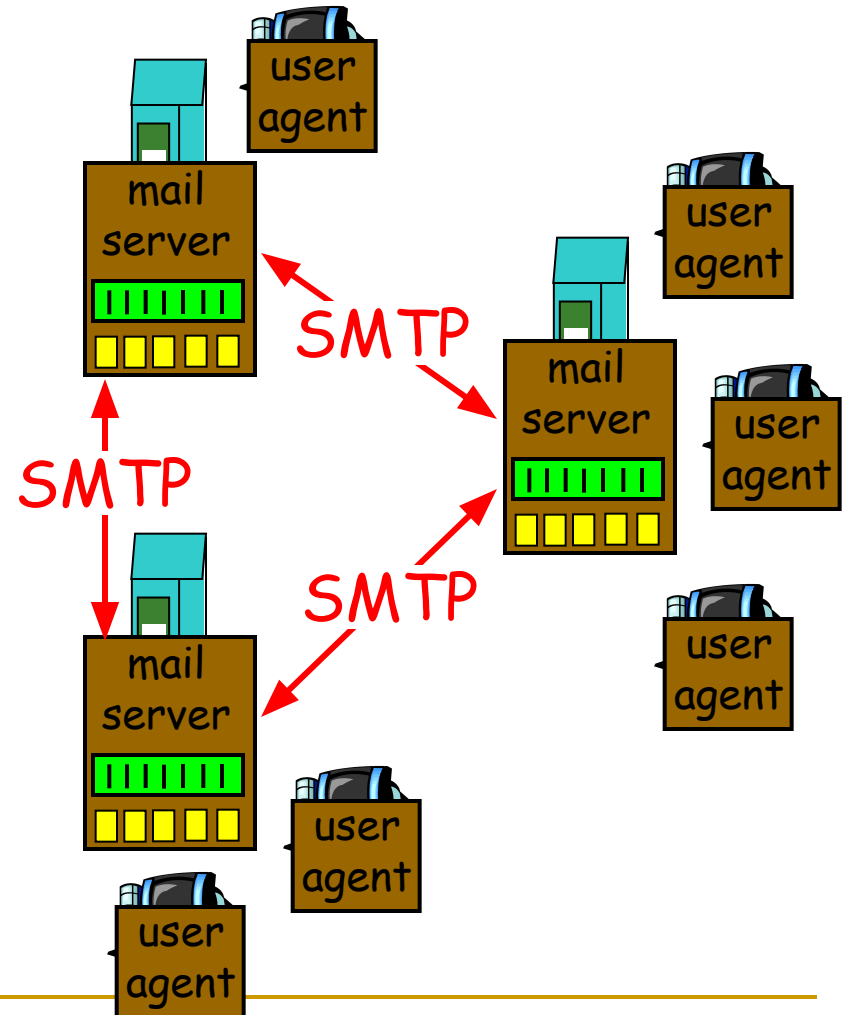
Services of User Agent



Electronic Mail: mail servers

Mail Servers

- **mailbox** contains incoming messages for user
- **message queue** of outgoing (to be sent) mail messages
- **SMTP protocol** between mail servers to send email messages
 - client: sending mail server
 - “server”: receiving mail server

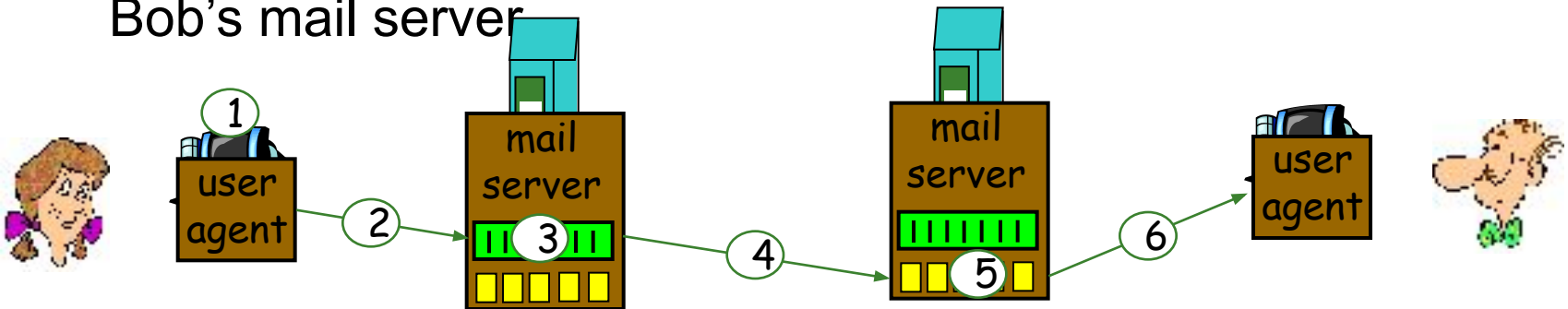


Electronic Mail: SMTP [RFC 2821]

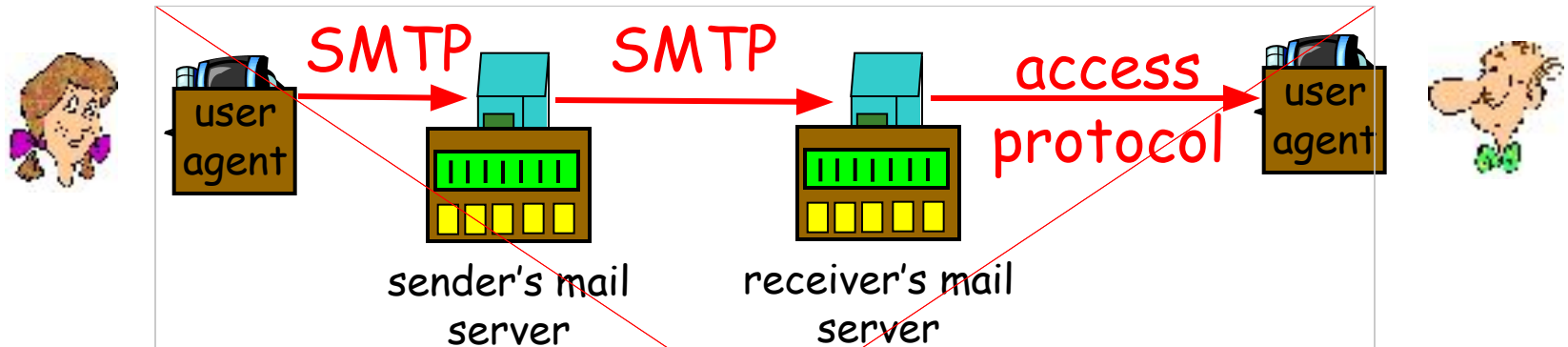
- uses TCP to reliably transfer email message from client to server, port 25
- direct transfer: sending server to receiving server
- three phases of transfer
 - handshaking (greeting)
 - transfer of messages
 - closure
- messages must be in 7-bit ASCII

Scenario: Alice sends message to Bob

- 1) Alice uses UA to compose message and “to”
`bob@someschool.edu`
- 2) Alice’s UA sends message to her mail server;
message placed in message queue
- 3) Client side of SMTP opens TCP connection with Bob’s mail server
- 4) SMTP client sends Alice’s message over the TCP connection
- 5) Bob’s mail server places the message in Bob’s mailbox
- 6) Bob invokes his user agent to read message

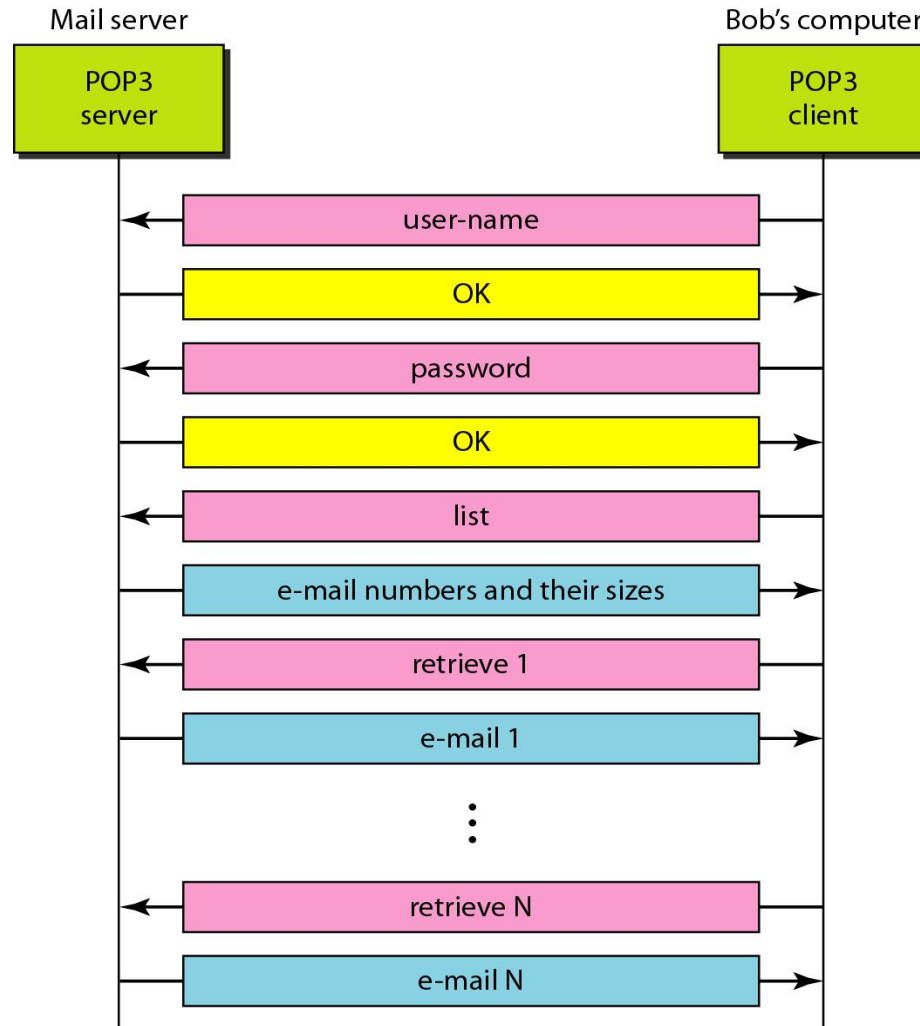


Mail access protocols

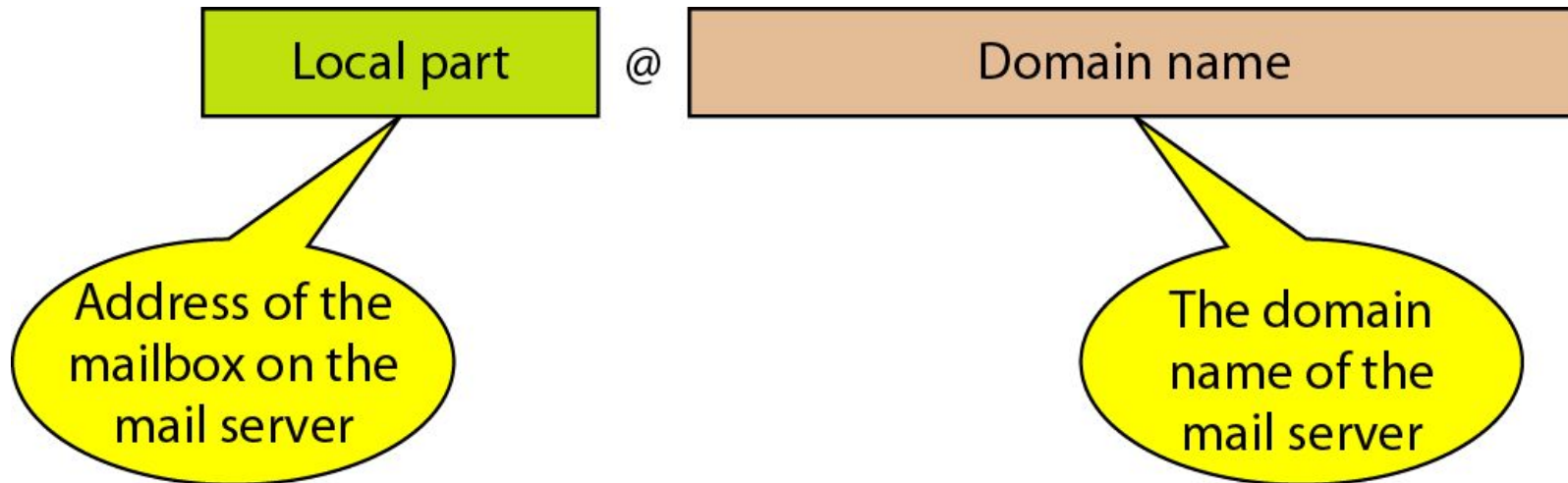


- SMTP: delivery/storage to receiver's server
- Mail access protocol: retrieval from server
 - POP: Post Office Protocol [RFC 1939]
 - authorization (agent <-->server) and download
 - IMAP: Internet Mail Access Protocol [RFC 1730]
 - more features (more complex)
 - manipulation of stored msgs on server
 - HTTP: Hotmail , Yahoo! Mail, etc.

The exchange of commands and responses in POP3

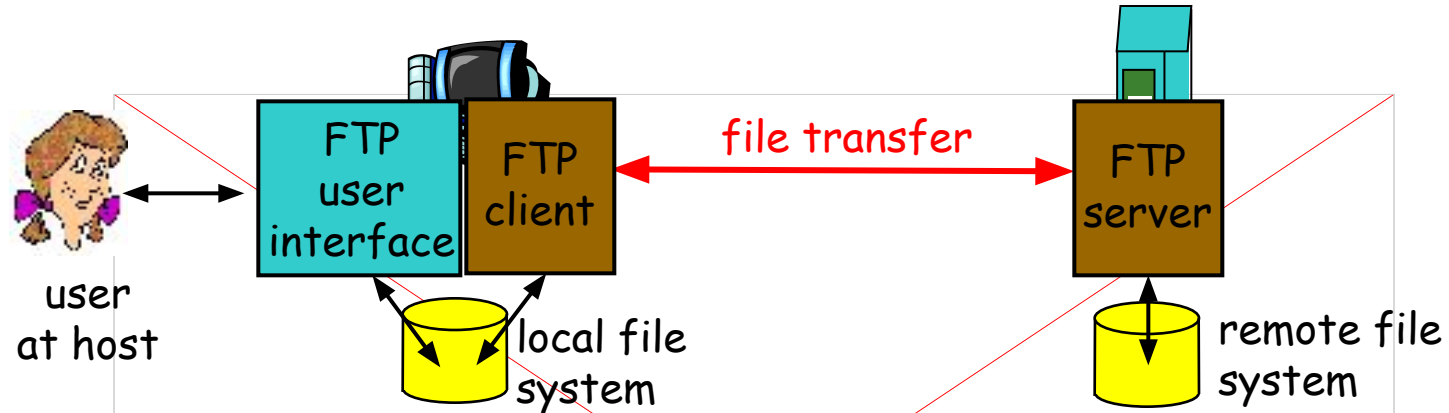


E-mail address



FTP

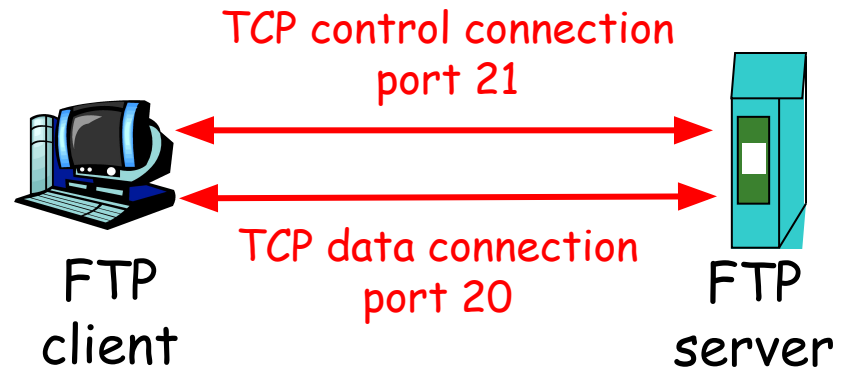
FTP: the file transfer protocol



- transfer file to/from remote host
- client/server model
 - *client*: side that initiates transfer (either to/from remote)
 - *server*: remote host
- ftp: RFC 959
- ftp server: port 21

FTP: separate control, data connections

- FTP client contacts FTP server at port 21, specifying TCP as transport protocol
- Client obtains authorization over control connection
- Client browses remote directory by sending commands over control connection.
- When server receives a command for a file transfer, the server opens a TCP data connection to client
- After transferring one file, server closes connection.



- Server opens a second TCP data connection to transfer another file.
- Control connection: “out of band”
- FTP server maintains “state”: current directory, earlier authentication